Problems on Ideals and Fields

From the class book: Chapter 15, Section 15.4; Chapter 16, Section 16.5; Chapter 19; Chapter 22, Sections 22.1 and 22.2

1. Key definitions

Sum, product, intersection.

- For ideals I, J of a ring R: $I + J = \{a + b : a \in I,\ b \in J\}$ is the smallest ideal containing $I \cup J$; IJ is the set of finite sums $\sum a_k b_k$ with $a_k \in I,\ b_k \in J$; $I \cap J$ is an ideal and $IJ \subseteq I \cap J$.
- The union $I \cup J$ is in general *not* an ideal.

Comaximal ideals.

- Ideals I, J of a commutative ring with unity (CRU) are comaximal if $I + J = R$.
- Then $IJ = I \cap J$ and $R/IJ \cong R/I \times R/J$ (Chinese Remainder Theorem).

Annihilator and radical.

- $\mathrm{Ann}(a) = \{x : xa = 0\}$; $\sqrt{I} = \{x : x^n \in I \text{ for some } n\}$.
- In $\mathbb{Z}$, $\sqrt{n\mathbb{Z}} = \mathrm{rad}(n)\mathbb{Z}$, where $\mathrm{rad}(n)$ is the product of the distinct primes dividing n.

Prime and maximal ideals.

- In a CRU, I is maximal iff R/I is a field, and prime iff R/I is an integral domain.
- Maximal implies prime (in a CRU).

PID.

- An integral domain in which every ideal is principal.
- A CRU (not necessarily a domain) with all ideals principal is a principal ideal ring, e.g. $\mathbb{Z}_n$.

Galois field. $\mathbb{F}_{p^n} = GF(p^n)$: the unique (up to isomorphism) field with p^n elements, the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$.

2. Main results

Theorem (Operations on ideals of $\mathbb{Z}$, and of any PID)

$$m\mathbb{Z} + n\mathbb{Z} = \gcd(m,n)\mathbb{Z}, \quad m\mathbb{Z} \cap n\mathbb{Z} = \mathrm{lcm}(m,n)\mathbb{Z}, \quad (m\mathbb{Z})(n\mathbb{Z}) = mn\mathbb{Z}.$$

- Hence $m\mathbb{Z} \cap n\mathbb{Z} = (m\mathbb{Z})(n\mathbb{Z})$ iff $\gcd(m,n) = 1$.
- In a PID the same holds with $\langle a \rangle, \langle b \rangle$.

Theorem (Counting ideals)

- (i) If $n = p_1^{a_1} \cdots p_k^{a_k}$, then $\mathbb{Z}_n$ has $\tau(n) = (a_1+1)\cdots(a_k+1)$ ideals, one $\langle d \rangle$ for each $d \mid n$.
- (ii) Ideals of $R_1 \times R_2$ (rings with unity) are exactly $I_1 \times I_2$, so counts multiply.
- (iii) $F[x]/\langle g \rangle$ has as many ideals as g has monic divisors; for $g = f_1^{e_1} \cdots f_r^{e_r}$ (distinct monic irreducibles) this is $(e_1+1)\cdots(e_r+1)$.
- (iv) $M_n(R)$: two-sided ideals are $M_n(I)$, I an ideal of R; left ideals of $M_n(F)$ correspond to subspaces of F^n.

Theorem (Counting prime and maximal ideals)

- $\mathbb{Z}_n$ has exactly $\omega(n)$ prime ideals, all maximal (ω = number of distinct prime factors).
- $\mathbb{Z}_m \times \mathbb{Z}_n$ has $\omega(m) + \omega(n)$ maximal ideals.
- $F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle$ has exactly r maximal ideals and has no non-zero nilpotent iff every $e_i = 1$.
- In a finite CRU every prime ideal is maximal.

Theorem (PIDs)

- In a PID every non-zero prime ideal is maximal, and $\langle p \rangle$ is maximal iff p is irreducible iff p is prime.
- Every PID is Noetherian and a UFD.
- $R[x]$ is a PID iff R is a field. A finite PID is a field.
- *Proof idea for the last-but-one:* $\langle x \rangle$ is a non-zero prime ideal of $R[x]$, so it is maximal and $R \cong R[x]/\langle x \rangle$ is a field.

Theorem (Structure of finite fields)

- A finite field F has prime characteristic p, contains $\mathbb{F}_p$, and $|F| = p^n$ with $n = [F : \mathbb{F}_p]$.
- For each p^n there is exactly one such field up to isomorphism.
- If $f \in \mathbb{F}_p[x]$ is irreducible of degree n then $\mathbb{F}_p[x]/\langle f \rangle \cong \mathbb{F}_{p^n}$.
- The group $F^\times$ is cyclic of order $q-1$, and $a^q = a$ for all $a \in \mathbb{F}_q$, i.e. $x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$.

Theorem (Subfields)

- $\mathbb{F}_{p^n}$ has a subfield of order p^m iff $m \mid n$; it is unique, namely $\{a : a^{p^m} = a\}$.
- So $\mathbb{F}_{p^n}$ has exactly $\tau(n)$ subfields, and $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n} \iff m \mid n$.

3. Standard quotients and formula tables

Quotient	Order	Field iff	Integral domain iff
$\mathbb{Z}/n\mathbb{Z}$	n	n prime	n prime (or $n = 0$)
$F[x]/\langle f \rangle$	$\lvert F \rvert^{\deg f}$	f irreducible	f irreducible
$\mathbb{Z}[x]/\langle p, f \rangle \cong \mathbb{Z}_p[x]/\langle \bar{f} \rangle$	$p^{\deg f}$	$\bar{f}$ irreducible mod p	same
$\mathbb{Z}[x]/\langle f \rangle$, $\deg f \geq 1$	infinite	never	f irreducible in $\mathbb{Z}[x]$
$\mathbb{Z}[i]/\langle n \rangle$, $n \geq 2$	n^2	n prime, $n \equiv 3 \pmod 4$	same
$\mathbb{Z}[i]/\langle a + bi \rangle$	$a^2 + b^2$	$a + bi$ a Gaussian prime	same

Ring	Number of ideals	Example
field; $M_n(F)$ (two-sided)	2	$\mathbb{Q}$, $M_2(\mathbb{R})$
$\mathbb{Z}_n$	$\tau(n)$	$\mathbb{Z}_{36}$: 9
$\mathbb{Z}_m \times \mathbb{Z}_n$	$\tau(m)\tau(n)$	$\mathbb{Z}_{12} \times \mathbb{Z}_8$: 24
$F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle$	$\prod(e_i + 1)$	$\mathbb{Q}[x]/\langle x^3 - x \rangle$: 8
$M_n(\mathbb{Z}_m)$ (two-sided)	$\tau(m)$	$M_2(\mathbb{Z}_6)$: 4
left ideals of $M_2(\mathbb{F}_q)$	$q + 3$	$M_2(\mathbb{Z}_2)$: 5

Quantity for $\mathbb{F}_q$, $q = p^n$	Formula	For $\mathbb{F}_{64}$
characteristic	p	2
order of $\mathbb{F}_q^\times$	$q - 1$	63
generators of $\mathbb{F}_q^\times$ (primitive elements)	$\varphi(q - 1)$	36
elements of order k ($k \mid q - 1$)	$\varphi(k)$	order 7: 6
solutions of $x^k = 1$	$\gcd(k, q - 1)$	$x^9 = 1$: 9
number of subfields	$\tau(n)$	4
$\lvert \mathrm{Aut}(\mathbb{F}_{p^n}) \rvert$	n (cyclic, Frobenius)	6
α with $\mathbb{F}_p(\alpha) = \mathbb{F}_{p^n}$	$nN_p(n)$	54
non-zero squares	$(q - 1)/2$ for odd q; all for even q	all 63

- As groups, $(\mathbb{F}_{p^n}, +) \cong \mathbb{Z}_p^n$ (not cyclic for $n \geq 2$) while $\mathbb{F}_{p^n}^\times \cong \mathbb{Z}_{p^n - 1}$.
- For odd q, -1 is a square in $\mathbb{F}_q$ iff $q \equiv 1 \pmod 4$.

4. Exam shortcuts and standard counter-examples

Exam shortcut.

- To decide prime/maximal, compute R/I: a field gives maximal; a domain that is not a field gives prime but not maximal; zero divisors give not prime.
- In $\mathbb{Z}[x]$ the maximal ideals are exactly $\langle p, f\rangle$ with f irreducible mod p.

Exam shortcut.

- To show an ideal is the whole ring, write 1 as a combination of generators, or find a unit in it.
- In $\mathbb{Z}[i]$, an ideal containing two integers m, n with $\gcd(m,n) = 1$ is all of $\mathbb{Z}[i]$.

Exam shortcut.

- If $\gcd(a,b) = 1$ then $\mathbb{Z}[i]/\langle a+bi\rangle \cong \mathbb{Z}_{a^2+b^2}$ (the image of $\mathbb{Z}$ already has a^2+b^2 elements).
- So it is a field iff a^2+b^2 is prime.

Standard counter-examples.

- Prime but not maximal: $\langle 0\rangle$ in $\mathbb{Z}$; $\langle x\rangle$ and $\langle x^2+1\rangle$ in $\mathbb{Z}[x]$.
- Maximal but not prime (no unity): $4\mathbb{Z}$ in $2\mathbb{Z}$.
- $\langle x^2+1\rangle$ is maximal in $\mathbb{R}[x]$, not prime in $\mathbb{C}[x]$, prime but not maximal in $\mathbb{Z}[x]$.
- Non-principal ideals: $\langle 2, x\rangle \subset \mathbb{Z}[x]$, $\langle x, y\rangle \subset F[x,y]$, $\langle 2, 1+\sqrt{-5}\rangle \subset \mathbb{Z}[\sqrt{-5}]$.
- UFD not PID: $\mathbb{Z}[x]$, $F[x,y]$. PID not Euclidean: $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$. Subring of a PID that is not a PID: $\mathbb{Z}[2i] \subset \mathbb{Z}[i]$.
- The diagonal $\{(a,a)\}$ is a subring of $\mathbb{Z}\times\mathbb{Z}$ not of the form $S_1 \times S_2$ and not an ideal.
- $\mathbb{F}_8 \not\subseteq \mathbb{F}_{16}$: containment needs $3 \mid 4$, not $8 < 16$.
- A finite field is never algebraically closed: $1 + \prod_{a\in F}(x-a)$ has no root.

5. Solved examples

Example 1. In $\mathbb{Z}$, find $I+J$, $I\cap J$, IJ for $I = 12\mathbb{Z}$, $J = 18\mathbb{Z}$. Find $\sqrt{72\mathbb{Z}}$ and the number of nilpotent elements of $\mathbb{Z}_{72}$.

Solution

- $I+J = \gcd(12,18)\mathbb{Z} = 6\mathbb{Z}$, $I\cap J = 36\mathbb{Z}$, $IJ = 216\mathbb{Z}$; since $\gcd \neq 1$, $IJ \subsetneq I\cap J$.
- As $72 = 2^3\cdot 3^2$, $\sqrt{72\mathbb{Z}} = 6\mathbb{Z}$.
- An element $a \in \mathbb{Z}_{72}$ is nilpotent iff $6 \mid a$, giving $72/6 = 12$ nilpotents (including 0).

Example 2. Find the number of ideals, maximal ideals and prime ideals of $R = \mathbb{Z}_{36} \times \mathbb{Z}_{10}$.

Solution

- Ideals are $I_1 \times I_2$: $\tau(36)\tau(10) = 9 \cdot 4 = 36$.
- Maximal ideals are $M \times \mathbb{Z}_{10}$ or $\mathbb{Z}_{36} \times M$ with M maximal: $\omega(36) + \omega(10) = 2 + 2 = 4$, namely $2\mathbb{Z}_{36} \times \mathbb{Z}_{10}$, $3\mathbb{Z}_{36} \times \mathbb{Z}_{10}$, $\mathbb{Z}_{36} \times 2\mathbb{Z}_{10}$, $\mathbb{Z}_{36} \times 5\mathbb{Z}_{10}$.
- R is a finite commutative ring with unity, so prime = maximal: 4 prime ideals.

Example 3. Compare $A = \mathbb{Z}_5[x]/\langle x^4 - 1\rangle$, $B = \mathbb{Z}_2[x]/\langle x^4 - 1\rangle$, $C = \mathbb{Z}_5[x]/\langle x^4 + 1\rangle$: number of ideals, maximal ideals, and whether each is reduced (no non-zero nilpotent).

Solution

- Over $\mathbb{Z}_5$, every non-zero element satisfies $a^4 = 1$, so $x^4 - 1 = (x-1)(x-2)(x-3)(x-4)$ and $A \cong \mathbb{Z}_5^4$: $2^4 = 16$ ideals, 4 maximal, reduced.
- Over $\mathbb{Z}_2$, $x^4 - 1 = (x+1)^4$: monic divisors $(x+1)^k$, $0 \le k \le 4$, so B has 5 ideals, a unique maximal ideal $\langle x+1\rangle$, and $x+1$ is a non-zero nilpotent; B is local and not reduced.
- Over $\mathbb{Z}_5$, $x^4 + 1$ has no root ($a^4 = 1 \neq -1$), but $x^4 + 1 = (x^2+2)(x^2-2)$, both irreducible since ± 2 are non-squares mod 5.
- So $C \cong \mathbb{F}_{25} \times \mathbb{F}_{25}$: 4 ideals, 2 maximal, reduced.

Example 4. Classify as maximal / prime but not maximal / not prime in $\mathbb{Z}[x]$: (a) $\langle 3, x^2 + x + 2\rangle$ (b) $\langle 5, x^2 + 1\rangle$ (c) $\langle x^2 + 1\rangle$ (d) $\langle 2, x^2 + 1\rangle$ (e) $\langle x\rangle$.

Solution

- (a) $\mathbb{Z}[x]/\langle 3, f\rangle \cong \mathbb{Z}_3[x]/\langle x^2 + x + 2\rangle$; values at 0, 1, 2 are $2, 1, 2$, no root, so irreducible: a field of order 9, **maximal**.
- (b) Mod 5, $x^2 + 1 = (x-2)(x-3)$, quotient $\cong \mathbb{Z}_5 \times \mathbb{Z}_5$: **not prime**.
- (c) Quotient $\cong \mathbb{Z}[i]$, a domain but not a field: **prime, not maximal**.
- (d) Mod 2, $x^2 + 1 = (x+1)^2$, quotient $\mathbb{Z}_2[x]/\langle (x+1)^2\rangle$ has a nilpotent: **not prime**.
- (e) Quotient $\cong \mathbb{Z}$: **prime, not maximal** (contained in $\langle 2, x\rangle$).

Example 5. In $\mathbb{Z}[i]$: (a) Show $\langle 3, 1+2i\rangle = \mathbb{Z}[i]$. (b) Identify $\mathbb{Z}[i]/\langle 3\rangle$, $\mathbb{Z}[i]/\langle 5\rangle$, $\mathbb{Z}[i]/\langle 1+3i\rangle$ and $\mathbb{Z}[i]/\langle 2+i\rangle$.

Solution

- (a) The ideal contains $9 = 3\cdot 3$ and $5 = (1+2i)(1-2i)$, so it contains $1 = 2\cdot 5 - 9$.
- (b) $3 \equiv 3 \pmod 4$ is a Gaussian prime: $\mathbb{Z}[i]/\langle 3\rangle \cong \mathbb{F}_9$, a field.
- $5 = (2+i)(2-i)$ with comaximal factors, so $\mathbb{Z}[i]/\langle 5\rangle \cong \mathbb{Z}_5 \times \mathbb{Z}_5$ (not a domain).
- For $1+3i$: $\gcd(1,3) = 1$, so the quotient is $\mathbb{Z}_{10}$, not a domain; indeed $1+3i = (1+i)(2+i)$.
- For $2+i$: norm 5 is prime, quotient $\cong \mathbb{Z}_5$, a field.

Example 6. Let $F = \mathbb{F}_{2^{12}}$. (a) List its subfields and decide whether $\mathbb{F}_8 \subseteq \mathbb{F}_{16}$. (b) What is the smallest subfield containing both $\mathbb{F}_{16}$ and $\mathbb{F}_{64}$, and what is $\mathbb{F}_{16} \cap \mathbb{F}_{64}$? (c) For $\mathbb{F}_{64}$, find the number of primitive elements and of α with $\mathbb{F}_2(\alpha) = \mathbb{F}_{64}$.

Solution

- (a) Subfields $\mathbb{F}_{2^d}$, $d \mid 12$: $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8, \mathbb{F}_{16}, \mathbb{F}_{64}, \mathbb{F}_{2^{12}}$ (six).
- $\mathbb{F}_8 \not\subseteq \mathbb{F}_{16}$ since $3 \nmid 4$.
- (b) The compositum is $\mathbb{F}_{2^{\mathrm{lcm}(4,6)}} = \mathbb{F}_{2^{12}}$ and the intersection is $\mathbb{F}_{2^{\gcd(4,6)}} = \mathbb{F}_4$.
- (c) $\mathbb{F}_{64}^\times \cong \mathbb{Z}_{63}$ has $\varphi(63) = 36$ generators.
- Elements in no proper subfield: proper subfields are $\mathbb{F}_8, \mathbb{F}_4$ (and $\mathbb{F}_2 \subset$ both), with $\mathbb{F}_8 \cap \mathbb{F}_4 = \mathbb{F}_2$, so $64 - (8+4-2) = 54 = 6\cdot N_2(6) = 6\cdot 9$.
- Note $36 < 54$: every primitive element generates the field, not conversely.